

Nucleic Acids

Lecturer 4

Presented By

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Chapter III

DNA

Chapter Three: DNA

- 1- DNA base composition
- 2- Secondary structure of DNA
- 3- A, B, and Z forms of DNA
- 4- Packing DNA into chromosomes

1- DNA base composition

- DNA has: Equal numbers of adenine and thymine residues ($A=T$) & Equal numbers of guanine and cytosine residues ($G=C$).
- These relationships, known as **Chargaff's rules**, were discovered in the late 1940s by Erwin Chargaff, who devised the first reliable quantitative methods for the compositional analysis of DNA.



Adenine = Thymine

Guanine = Cytosine

Chargaff's Rule

1- DNA base varies among diff. organism

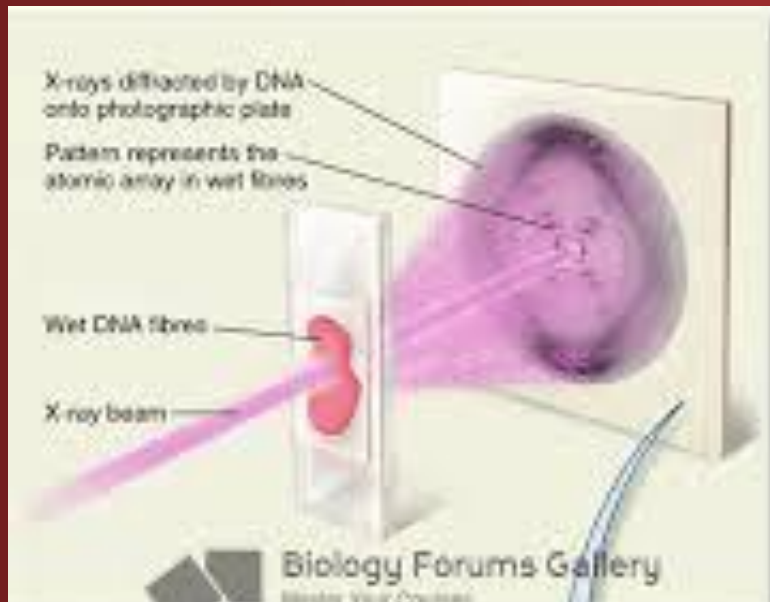
- In different bacteria species it ranges from 25 to 75 % G+C .
- However, it is more or less constant among related species; for example, in mammals G+C ranges from 39 to 46%.

Table 5.1: Percent Composition of Each Base from DNA of Several Species

Organism	Adenine	Thymine	Guanine	Cytosine
<i>Mycobacterium tuberculosis</i>	15.1	14.6	34.9	35.4
<i>Escherichia coli</i>	26.0	23.9	24.9	25.2
Yeast	31.3	32.9	18.7	17.1
<i>Drosophila melanogaster</i>	27.3	27.6	22.5	22.5
Mouse	29.2	29.4	21.7	19.7
Human (liver)	30.7	31.2	19.3	18.8

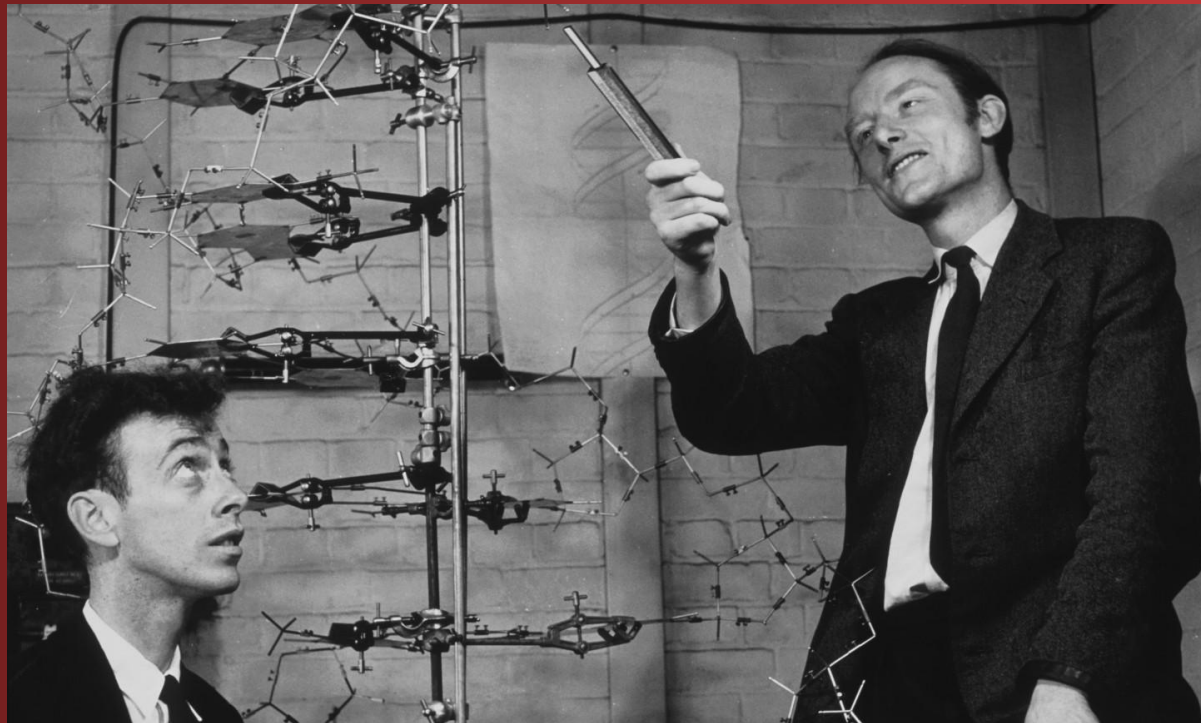
2-The Secondary Structure of DNA

- X-ray diffraction involves taking a highly concentrated solution of DNA and stretching it into a fiber that results in the orientation of the molecules.



2-The Secondary Structure of DNA

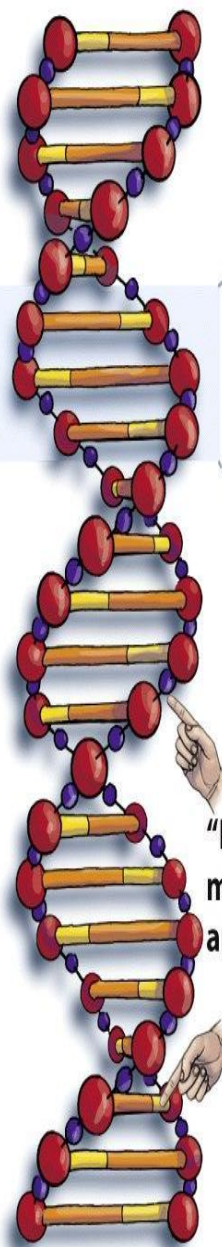
- Using early information obtained by this X-Ray diffraction, coupled with chemical analysis and model building, Watson and Crick in 1953 proposed that the DNA molecule extended chain having a highly ordered structure composed of:



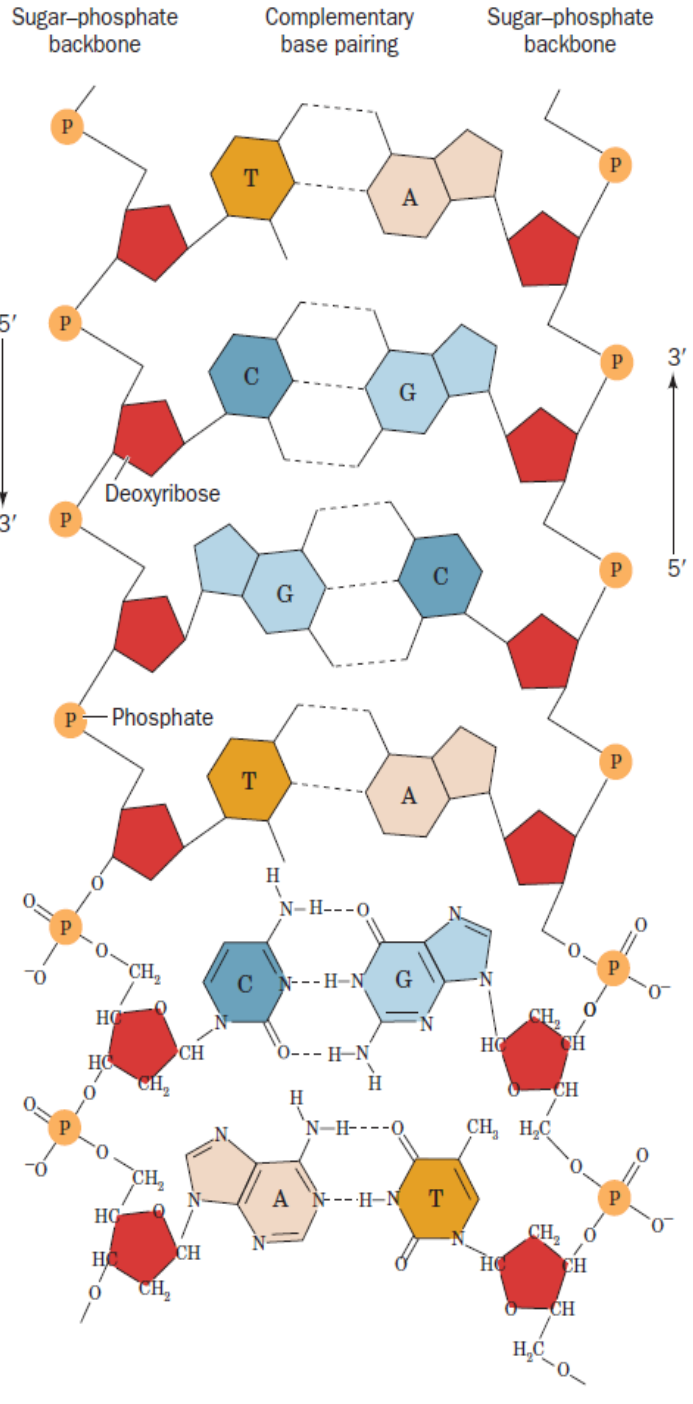
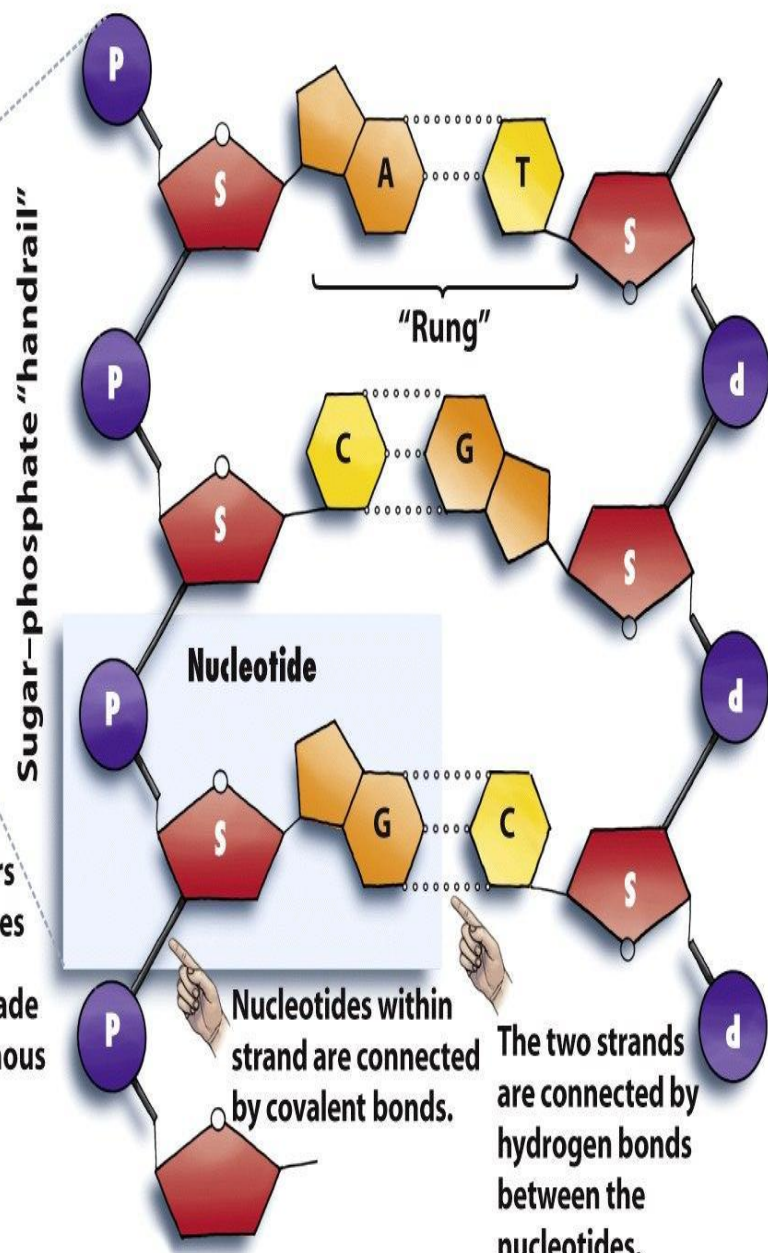
2-The Secondary Structure of DNA

- The two strands of DNA are arranged anti-parallel to one another: viewed from left to right the "top" strand is aligned 5' to 3', while the "bottom" strand is aligned 3' to 5'. This is always the case for duplex nucleic acids.
- G-C base pairs have 3 hydrogen bonds, whereas A-T base pairs have 2 hydrogen bonds: one consequence of this disparity is that it takes more energy (e.g. a higher temperature) to disrupt GC-rich DNA than AT-rich DNA.
- Two helical DNA chains wound around the same axis to form a right-handed double helix anti-parallel strand.

DNA double helix is made of two strands.



Each strand is a chain of of antiparallel nucleotides.



2-The Secondary Structure of DNA

- Hydrophilic backbones of alternating deoxyribose and phosphate groups; faces watery environment.
- Bases are stacked flat and perpendicular to the axis. There are 10 base pairs per helical turn.
- The surface of DNA displays a major and minor groove
- H bonding between the bases of the complementary strands in which two linear strands of DNA are wound around one another.
- The major force promoting formation of this helix is complementary base pairing: Adenine form hydrogen bonds with Thymine (or Uracil in RNA), and Guanine form hydrogen bonds with Cytosine.
- The DNA strands can be separated by Melting (Heat or low salt) and re-annealing (Cool or high salt)

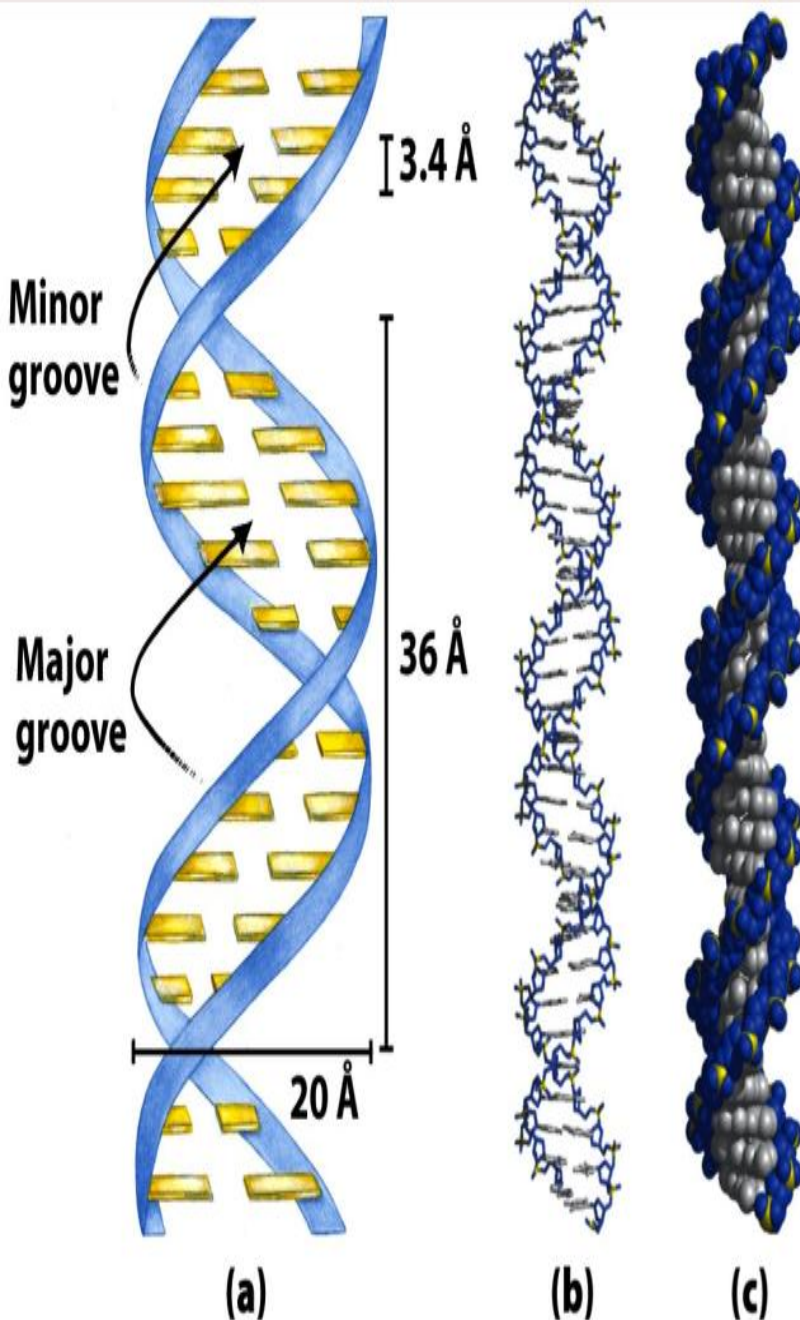


Figure 8-13
Lehninger Principles of Biochemistry, Fifth Edition
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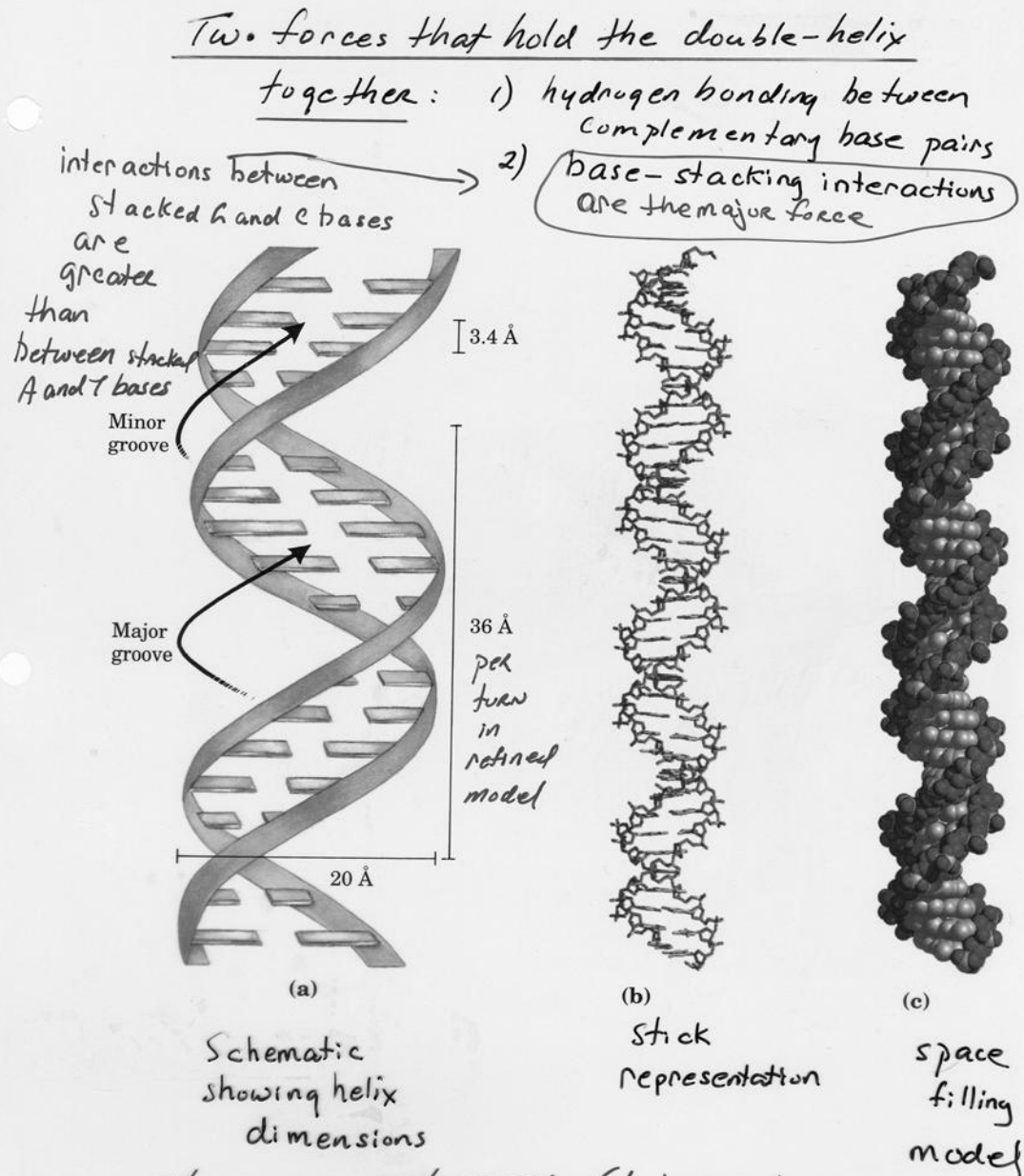
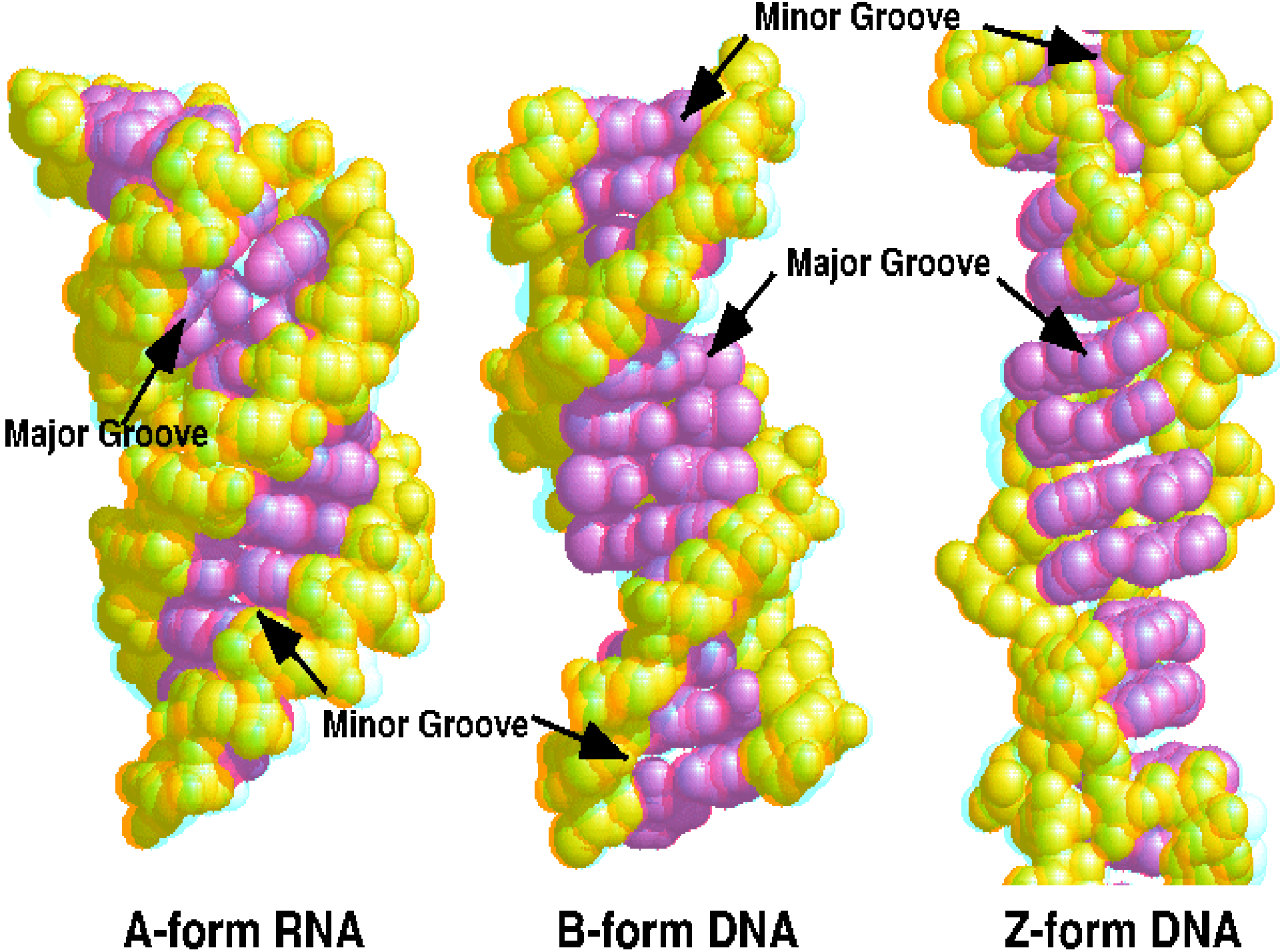


figure 10-15
Nelson and Cox: Lehninger Principles of Biochemistry, Third Edition
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3- A, B, and Z forms of DNA

- Wilkins and his colleagues demonstrated that, depending on the conditions chosen to produce the DNA fibers, they can have a variety of possible conformations (structures).
- **B-form DNA:** represent the conformation of most DNA found in cells. x-ray diffraction data of DNA fibers at high humidity (92%).
- **A-form DNA:** predominant conformation of DNA-RNA hybrids and double stranded RNA segments. X-ray fiber diffraction studies at low humidity (75%).
- **Z-form DNA:** Very minor amounts of Z-DNA exists in-vivo. DNA containing alternating purine and pyrimidine residues (dinucleotides: CGCGCGCG) can fold up into left-handed as well as right handed helices. Found under conditions of high conc. of positively charged ions (e.g Na⁺).

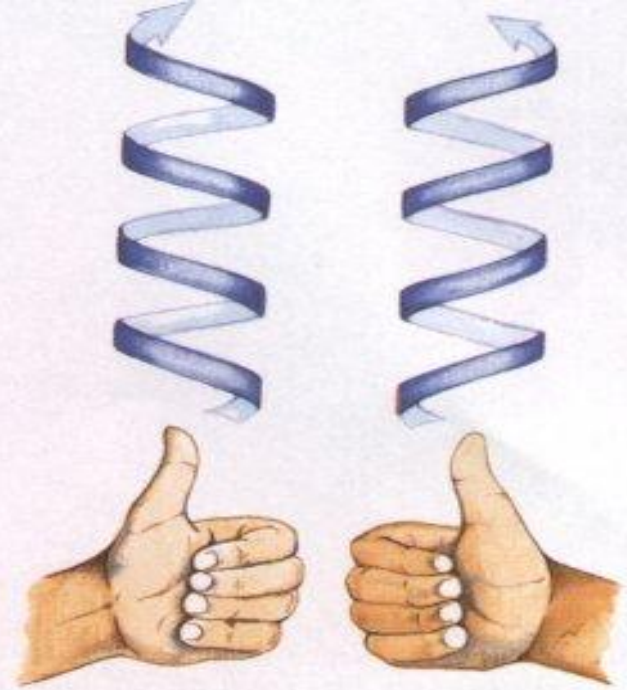
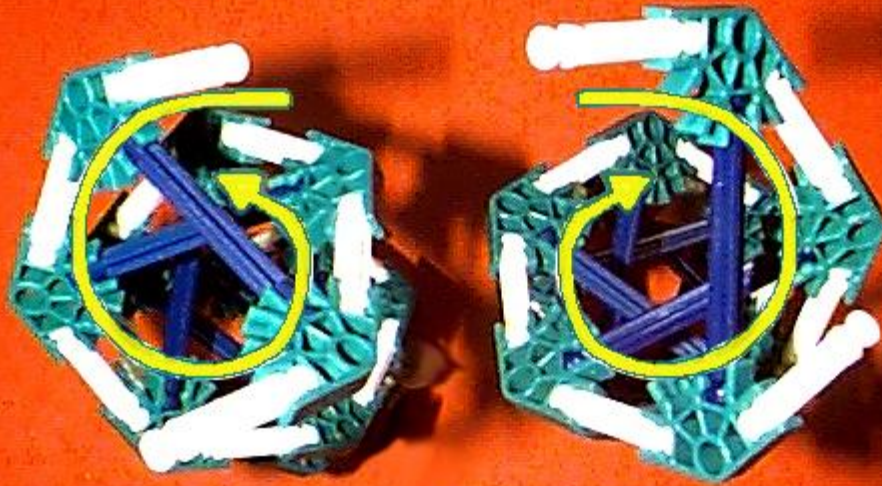


3-Differences between A, B, and Z DNA

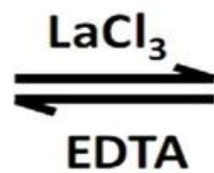
Feature	B-DNA	A-DNA	Z-DNA
Type of helix	Right-handed	Right-handed	Left-handed
Helical diameter (nm)	2.37	2.55	1.84
Rise per base pair (nm)	0.34	0.29	0.37
Distance per complete turn (pitch) (nm)	3.4	3.2	4.5
Number of base pairs per complete turn	10	11	12
Topology of major groove	Wide, deep	Narrow, deep	Flat
Topology of minor groove	Narrow, shallow	Broad, shallow	Narrow, deep

Left-Handed
Helix

Right-Handed
Helix



B-DNA
Right handed



Z-DNA
Left handed

